

Mobile Number:
+254-713793373
+254-737766052

Address: P.O. Box 450 –
90200, Kitui

Email Address: faithmueni24@gmail.com
mueni.faith@embuni.ac.ke

Status: Married

Dr. Faith Mueni Musyoka

SENIOR LECTURER / COMPUTING / RESEARCHER / GRANT-WRITER

Google Scholar: <https://scholar.google.com/citations?hl=en&user=I8-9ixYAAAAJ>

ORCID Account: <https://orcid.org/0000-0002-9574-8235>

Linkedin: <https://www.linkedin.com/in/dr-faith-m-4ab5b133/>

Personal Profile

Senior Lecturer in Computing and Information Technology, researcher, and grant funded project leader with strong expertise in Artificial Intelligence, Data Governance, Health Informatics, and Digital Systems. Experienced in academic leadership, curriculum development, and multi institution research collaboration. Proven record in funded AI and digital innovation projects, publications, and postgraduate supervision. Passionate about responsible AI, ethical technology use, and capacity building for students and youth in emerging digital fields. Recognized for combining technical knowledge, policy awareness, and practical implementation in low resource and higher education environments.

Career Objective

To advance as a leader and consultant in Artificial Intelligence, Data Governance, and Responsible Digital Innovation, contributing to research, policy informed practice, and institutional transformation. To support organizations and communities through ethical AI adoption, capacity building, and evidence based digital solutions while growing in academic, policy, and global AI governance spaces.

Personal Affiliation

2014 – To Date	Association of Computing for Machinery (ACM). Full Member. No: 3947008
2023 – To Date	Professional member of Organization for Women in Science for the Developing World (OSWD) Full Member. No: 14752
2024-To Date	Women in Cloud Network Full Member No: 27007463

Education

2015 - 2019	Doctor of Philosophy in Information Technology Kabarak University - Kenya Title: Internet of Things based Model for Preeclampsia Monitoring in Antenatal Care
2011 - 2014	Master of Science in Information Technology. Masinde Muliro University of Science and Technology – Kenya.
2007 - 2011	Bachelor of Science in Computer Science (Second Class Honors, Upper Division). Masinde Muliro University of Science and Technology – Kenya.

	2002-2005	Kenya certificate of Secondary Education (KCSE) Mbitini Girls' Secondary School Grade A – (minus)
Employment History	July 2025- To Date	Senior Lecturer, Computing & Information Technology University of Embu
	Dec 2019 – June 2025	Lecturer, Computing & Information Technology University of Embu
	Jan 2018 – Dec 2019	Tutorial Fellow, Mathematics, Computing and Informatics University of Embu
	Aug 2015 – Dec 2017	Assistant Lecturer, School of Computing and Informatics Gretsa University
	Sep 2014 – Aug 2015	Adjunct Lecturer Gretsa University
	January 2026- To Date	Adjunct Research Scientist African Institute for Capacity Development (AICAD)
	July 2025-To Date	Chair of Department (Computing & Information Technology) University of Embu • Provide academic and administrative leadership, coordinating teaching, examinations, and postgraduate supervision. • Oversee staff performance, mentorship, and professional development. • Manage departmental resources, budgets, and ensure compliance with University policies and quality standards. • Represent the department in committees, foster collaborations, and promote research, innovation, and community engagement.
University Administrative Assignments	June 2019 – July 2025	Coordinator, Website and E-Repository University of Embu As a Website & E-Repository Coordinator, I devised strategies that made the University rank and maintain top 10 in Webometrics ranking of Universities in January 2021 from position 50 in January 2019. In July 2025 the university website ranked position 6 in the country.
	April 2026-To Date	Chair- Committee on Development of Artificial Intelligence Guidelines University of Embu
	April 2026-To Date	Member- Professional and Short Courses Implementation Committee University of Embu
	April 2026- To Date	Member- Committee on Development of a framework for addressing incomplete student examination records University of Embu
	March 2026- To Date	Member- University Website Policy Development Committee University of Embu
	January 2026-To Date	Member- Practicum Placement Board University of Embu
	January 2026-To Date	Member- ISO Implementation Committee University of Embu

	September 2025-To Date	Member- Board of Postgraduate Studies Committee University of Embu
	November 2021-To Date	Secretariat, University of Embu partnership with Higher Education Partners (HEP) University of Embu
	January 2021-To Date	Member, Corporate Marketing & Publicity Committee University of Embu
	February 2021-To Date	Member, School of Pure and Applied Sciences Board of Postgraduate Studies University of Embu
	March 2021-To Date	Member, Adhoc Committee on Computer Literacy Skills Course Development and Implementation University of Embu
	November 2020-To Date	Secretary, Research Dissemination Workshop Planning Committee University of Embu
	November 2020-To Date	Programme Academic Advisor, Postgraduate programs University of Embu
	February 2020	Champion, Online Teaching and Learning Programme University of Embu
	Oct 2016 – Dec 2017	Head of School, Computing and Informatics Gretsa University
	28-30 July 2025	Research and Innovation Management (RIM) Project Second Annual Symposium in Azalai Hotel in Abidjan, gCote d'Ivoire <i>"Digital Intervention for the early detection and prevention of depression and suicide among Youth in Kenya"</i>
	9 th -12 th June 2025	4th Multisectoral Conference and Exhibition on Research, Science, Technology, and Innovation- NACOSTI <i>"Digital Intervention for the early detection and prevention of depression and suicide among Youth in Kenya"</i>
	9 th -11 th December 2024	Research and Innovation Management (RIM) Project First Annual Symposium in Speke Resort Munyonyo in Uganda. <i>"Digital Intervention for the early detection and prevention of depression and suicide among Youth in Kenya"</i>
	5 th -7 th November 2024	9th DeKUT International Conference on Science, Technology, Innovation & Entrepreneurship, Dedan Kimathi University of technology, Kenya Daniel Njiru, David Muchangi Mugo, Faith Mueni Musyoka: The Paradox of Knowledge Management Systems: Competency Traps, AI, and Organizational Adaptability
Conference and Workshops Participation	2 nd -3 rd Nov 2023	e-Learning Conference 2023 -University of Embu <i>"Quality Teaching and Learning, Assessment, Technology and Innovativeness in Virtual Environments"</i> . Faith Mueni Musyoka: Building a simulated mentorship platform for online learning programs
	8 th June 2023	National Research Fund. Capacity Building Workshop Programme R2c, Mental Wellness and Leather Value Chain Call for Proposal. KICC – Lenana Hall, Nairobi
	9 th – 10 th Dec 2021	International Conference on Electrical, Computer, and Energy Technologies (ICECET), Cape Town

		Dominic Kitavi Makaa, Faith Mueni Musyoka: Cramer-Rao Bound for Direction-of-arrival estimation using Triad of First-Order Cardioid Sensors
	8 th – 9 th Nov 2021	Moderator, 4th International Multidisciplinary Research Conference “Sustainability in the Era of Covid-19 Pandemic” 8 th –9 th November 2021 at Grets University
	25 th Oct 2018	Kabarak University 8th Annual International Research Conference: Translational and Preventive HealthCare. Faith Mueni Musyoka, Dr. Moses Thiga, Prof. Geoffrey Muchiri: A Blood Pressure Monitoring Prototype for Preeclampsia Management in Antenatal Care, Kenya.
	23 rd Jan 2018	University of Embu. Attended Training workshop on Effective Examination Setting, Administration, and Processing
	Nov 2017	International Multidisciplinary Conference Grets University: Faith Mueni Musyoka: User acceptance on mobile apps as an effective medium to learn C++ programming language: Conceptual study
	Nov 2017	International Multidisciplinary Conference Grets University: Beatrice Kairo, Faith Mueni Musyoka: Alert messaging system for Gatunguru Tea Factory Farmers in Murang’a County, Kenya
	Oct 2017	Kabarak University 7th Annual International Research Conference: Micro Research Workshop Faith Mueni Musyoka: SMS-based technology for maternal health outcome improvement in the informal settlement a case study of Rhonda slums, Nakuru County.

Professional Qualifications & trainings

March 2026	Training of Trainers: NITA & Corporate staffing Services Introduction to training, training needs analysis, design & implementation training programs, effective use of training aids, approaches to training, managing difficult learners, training evaluation
January 2025-December 2025:	Career Boost with Power BI DAX Language, Data Visualization, SQL, Advanced Ms. Excel, Power Automate, API Connections
4 th -25 th November 2025	Ethical and Responsible AI in Sub-Saharan Africa Training IDRC-CRDI, Hash Project, HealthAI
July 2024	Artificial Intelligence Fundamentals IBM SkillsBuild https://www.credly.com/badges/1c6daf41-2460-4e91-bbdf-0deb80f4a0a4
November 2023-March2024	Cybersecurity Fundamentals: Digital Regenesys Business School
November 2023-March2024	Cybersecurity Defence Toolbox: Digital Regenesys Business School
2021	Advanced Cyber Security Training for Teachers (Commonwealth of Learning)
2020	IBM SKILLS ACADEMY (Cloud Application Developer 2019 – Mastery Award)

Publications

1. Murikah, W., Nthenge, J. K., & **Musyoka, F. M.** (2026). Reconciling Cultural Values with Algorithmic Profiling Through Kenya's Data Protection Lens: A Conceptual Framework for Inferential Privacy. *Telematics and Informatics Reports*, 100318. <https://doi.org/10.1016/j.teler.2026.100318>
2. Wanyonyi, M., Morris, Z. N., **Musyoka, F. M.**, Makaa, D., & Kitavi, D. M. (2026). Balancing interpretability and accuracy: a comparative evaluation of logistic regression and machine learning approaches for coronary heart disease risk prediction in low-resource settings. *Authorea Preprints*.
3. Wanyonyi M, Morris ZN, **Musyoka FM**, Kitavi DM (2025) Enhanced machine learning and hybrid ensemble approaches for Coronary Heart Disease prediction. *PLoS One* 20(12): e0328338. <https://doi.org/10.1371/journal.pone.0328338>
4. Wambui, B. M., Kamau, J., & **Musyoka, F. M.** (2025). Enhancing Biometric Access Security in Pharmaceutical Companies through Contactless Multimodal Fusion: A Proposed Model Framework and Systematic Review. *Journal of Intelligent Systems and Applied Data Science*, 3(2). <https://doi.org/10.64680/jisads.v3i2.48>
5. Wanyonyi, M., Morris, Z. N., **Musyoka, F. M.**, & Kitavi, D. M. (2025). Predicting Coronary Heart Disease Using Classical Statistical Models: A Comparative Evaluation of Logistic Regression and Cox Proportional Hazards. *IAENG International Journal of Applied Mathematics*, 55(12).
6. Kogi, D., Mugo, D. M., **Musyoka, F. M.**, & Murikah, W. (2025). How Knowledge Management Systems Trap Dynamic Organisations into Outdated Practices: A Systematic Review. *Scientific African*, e02946. <https://doi.org/10.1016/j.sciaf.2025.e02946>
7. Njiru, D. K., Mugo, D. M., & **Musyoka, F. M.** (2025). Ethical Considerations in AI-Based User Profiling for Knowledge Management: A Critical Review. *Telematics and Informatics Reports*, 100205. <https://doi.org/10.1016/j.teler.2025.100205>
8. Mmbayi, I. I., Gakii, C., & **Musyoka, F. M.** (2025). Sentiment Analysis-Based Model for Monitoring User Engagement with Mental Health Chatbots. *Engineering Reports*, 7(6), e70247 doi: <https://onlinelibrary.wiley.com/doi/epdf/10.1002/eng2.70247>
9. Masinde, J. M., Mugambi, F., **Musyoka, F. M.**, & Masinde, R. C. (2025). Applying SERVQUAL Model in Library Service Delivery to Attain Students Satisfaction at a Private University in Kenya. *International Journal of Librarianship*, 10(1), 54-68. <https://doi.org/10.23974/ijol.2025.vol10.1.445>
10. Mboya, F. M., Wambugu, G. M., Oirere, A. M., Omuya, E. O., **Musyoka, F. M.**, & Gikandi, J. W. (2025). Enhancing Personalized Learning in Programming Education through Generative Artificial Intelligence Frameworks: A Systematic Literature Review. *International Journal of Advanced Trends in Computer Science and Engineering*, 14(2)
11. Mbevi, R. M., Kamau, J., & **Musyoka, F. M.** (2025). Content Based Approach for Detecting Smishing Messages in Mobile Phones Using an Improved Convolutional Neural Networks Model. *African Journal of Empirical Research*, 6(2), 188-204. <https://doi.org/10.51867/ajernet.6.2.17>
12. Fredrick Mugambi Muthengi, David Muchangi Mugo, Stephen Makau Mutua, Faith Mueni Musyoka. "A simplified approach to establishing the impact of software source code changes on requirements specifications". *Bulletin of Electrical Engineering and Informatics* (2025)
13. Wilberforce Murikah, Jeff Kimanga Nthenge & Faith Mueni Musyoka. "Bias and Ethics of AI systems in Auditing-A Systematic Review". *Scientific African* (2024), doi: <https://doi.org/10.1016/j.sciaf.2024.e02281>
14. Fredrick Mugambi Muthengi, David Muchangi Mugo, Stephen Makau Mutua, **Faith Mueni Musyoka**. "Method level static source code analysis on behavioral change impact analysis in software regression testing ". *Indonesian Journal of Electrical Engineering and Computer Science*. DOI: <https://doi.org/10.11591/ijeecs.v35.i1.pp665-672> . Vol. 35 (1), July 2024, pp. 665-672.
15. Duncan Kereu Kodhek, John Wachira Kamau, & **Faith Mueni Musyoka**. "Model for Total Cost Determination in Open-Source Software Ownership: Case of Kenyan Universities' Learning Management System". *Computer and Information Science*. Volume 17 Issue 1. March, 2024 [10.5539/cis.v17n1p18](https://doi.org/10.5539/cis.v17n1p18)

16. Nthenge, Jeff Kimanga and **Musyoka, Faith Mueni** and Mugo, David Muchangi. "Hybrid Machine Learning Algorithms for Risk Assessment in Insurance Industry: Empirical Review". *Heliyon* (2024). Doi: <http://dx.doi.org/10.2139/ssrn.4498613>
17. **Faith Mueni Musyoka**. "Role of a 24-hour Ambulatory Internet of Things System in Preeclampsia Monitoring: Technologies, Challenges, and Future Path Survey". *International Journal on Recent and Innovation Trends in Computing and Communication*. Volume 11 Issue 6. June, 2023: <https://doi.org/10.17762/ijritcc.v11i6.7387>
18. Jeff Kimanga, **Faith Mueni Musyoka** & David Mugo. "Implementing a Hybrid Model using K-Means Clustering and Artificial Neural Networks for Risk Prediction in Life Insurance". *International Journal on Recent and Innovation Trends in Computing and Communication*. Volume 11 Issue 7. Aug, 2023 <https://doi.org/10.17762/ijritcc.v11i9.9622>
19. Kitavi, D. M., & **Musyoka, F. M.** (2021, December). Cramer-Rao Bound for Direction-of-Arrival Estimation Using a Triad of First-Order Cardioid Sensors. In 2021 International Conference on Electrical, Computer and Energy Technologies (ICECET) (pp. 1-3). IEEE. 10.1109/ICECET52533.2021.9698456
20. Juliet Gathoni Muchori, Gabriel Kamau, **Faith Mueni Musyoka**. "Internet of Things based model for identifying paediatric emergency cases." *Advanced Medical Sciences: An International Journal (AMS)*. Vol 8, No.1/2/3, August 2021
21. **Faith Mueni Musyoka**. "Performance Evaluation of Internet of Things Prototype for Preeclampsia Monitoring in Antenatal Care". *International Journal of Engineering and Advanced Technology*. Volume 10 Issue 2. December 2020. <https://www.ijeat.org/wp-content/uploads/papers/v10i2/B20091210220.pdf>
22. **Faith Mueni Musyoka**, Moses Mwangi Thiga, Geoffrey Muchiri Muketha "A 24-hour ambulatory blood pressure monitoring system for preeclampsia management in antenatal care". *Informatics in Medicine-Unlocked*, Elsevier. Volume 16 2019. <https://doi.org/10.1016/j.imu.2019.100199>
23. **Faith Mueni Musyoka**, Moses Mwangi Thiga, Geoffrey Muchiri Muketha "An Assessment of Suitable and Affordable Smart armband for preeclampsia Management in Antenatal Care". *International Journal of Information Technology*, Volume 3, Issue 6, June, 2019
24. **Musyoka Faith Mueni** "Analysis of Static Code Techniques for Vulnerability Detection: A reviewed Literature" *International Journal of Modern Computer Science (IJMCS)*, ISSN: 2320-7868. Volume 5, Issue 5, October, 2017, pp 29-31
25. **Musyoka Faith Mueni** "Determinants of Mobile Apps in C++ Programming Course: A reviewed Literature" *International Journal of Modern Computer Science (IJMCS)*, ISSN: 2320-7868. Volume 5, Issue 5, October, 2017, pp 26-28
26. **Musyoka Faith Mueni**, Muketha Geoffrey Muchiri, Anselemo Peters Ikoha. "An Assessment framework for Internet banking system reliability" *International Journal of Technology in Computer Science & Engineering* (ISSN 2349-1582). Volume 1(3), September 2014, pp 88-100

Students' Supervision

1. **Graduated:** Juliet Gathoni muchori- Msc. Information Technology, Murang'a University of Technology: Internet of Things based model for identification of children emergency cases
2. **Graduated:** Jeff Kimanga- MSc. Information Technology, University of Embu: Implementation of a Hybrid Model using K-Means Clustering and ANN for Risk Prediction

3. **Graduated:** Kereu Kodhek Duncan- MSc. Information Technology, Mt. Kenya University, Kiambu: A Model for Total Cost Determination in Open-Source Software Ownership: Case of Kenyan Universities' Learning Management System.
4. **Graduated:** Fredrick Mugambi Muthengi – Ph.D. Computer Science, University of Embu: Software Regression Testing Performance Optimization Model
5. **Completed:** Daniel Kogi Njiru- Msc. Information Technology, University of Embu: Data Mining and User Profiling Framework for University Knowledge Management
6. **Completed:** Ian Igado- Msc. Information Technology, University of Embu: Sentiment analysis-based model for monitoring user engagement with mental health chatbots.
7. **Ongoing:** Maurice Wanyonyi- Phd in Statistics, University of Embu: Hybrid modeling approaches for predicting coronary heart disease: integrating classical, machine learning, and ensemble techniques
8. **Ongoing:** David Thuku- Msc. Information Technology, University of Embu: Feature selection and class balancing based model for breast cancer prediction
9. **Ongoing:** Philemon Kipngetchi- Msc. Information Technology, University of Embu: Development Model for Early Cardiovascular Disease Detection Using Retrieval Augmented Generation
10. **Ongoing:** Jonathan Saina- Msc. Information Technology, University of Embu: Blockchain Based Smart Contract Framework for Real Estate Records Management in Kenya
11. **Ongoing:** Boniface Mwangi Wambui- Phd in Information Technology, Mount Kenya University: Enhancing Biometric Access Level Security through Contactless Hybrid Fusion in Selected Pharmaceutical Firms in Nairobi County Kenya

Professional, Consulting, and Achievement

2026	Project Lead and Developer: University of Embu Counselling Service Platform. Co-led and directly participated in the design and development of a digital counselling service platform to support student mental health and wellbeing at the University of Embu. Contributed to system architecture, feature design, and implementation of core modules for confidential support access, guided self-help resources, and digital screening and referral workflows. Coordinated requirements gathering with counsellors and university units, and translated user needs into functional system components. Ensured integration of data privacy safeguards, ethical AI use principles, and responsible data governance practices. Collaborated with technical and health teams to support secure and scalable deployment.
2025	Project Lead-AI Ethics and Data Governance Youth Capacity Building Initiative Designed and launched a capacity building initiative focused on AI ethics and data governance for youth and students. Developed practical training modules on responsible AI use, data privacy, digital rights, bias awareness, and safe use of AI tools in learning and work environments. Delivering workshops and outreach sessions in partnership with youth and education institutions to promote responsible and accountable AI adoption.
2026	Curriculum Development- Bsc. Electrical and Electronic Engineering University of Embu
2024	Curriculum Developer for Msc. Information Technology & Msc. Computer science Machakos University
2022	Curriculum Developer for PhD. in Information Technology in University of Embu University of Embu

	2018	Curriculum Developer for Bsc. in Information Technology in University of Embu University of Embu
	2018	Curriculum Developer for Diploma in Information Technology in University of Embu University of Embu
	2018	Curriculum Developer for Certificate in Information Technology in University of Embu University of Embu
	2016	Curriculum Developer for Certificate in Information Technology in University of Embu Gretsa University
	2019 – To Date	Article Reviewer for academic manuscripts (Elsevier Journals: Informatics in Medicine Unlocked)
	2021	Curriculum Reviewer for Bsc. Computer Science Programme Jaramogi Odinga Oginga University
	July 2021 – To Date	Thesis Internal Examiner: School of Pure and Applied Sciences University of Embu I examined two projects
	March 2024- To Date	Thesis External Examiner: School of Pure and Applied Sciences Mount Kenya University
	June 2026- Date	Thesis External Examiner: School of Science and Technology USIU-Africa
	June 2026- Date	Thesis External Examiner: School of Computing Murang'a University of Technology
Research Grants Received	June 2026- Date	External Examiner: School of Informatics and Information Science Jaramogi Odinga Odinga University of Science and Technology
	Title	Africa AI Upskilling Programme for Higher Educational Institutions Role: PRINCIPAL INVESTIGATOR PERIOD: 5 th June 2026- 31 st January 2029 AWARDING BODY: FATE Foundation, African Institute for Mathematical Sciences (AIMS), and Google.org AMOUNT: USD 57,333.33
	Title	A digital intervention for the prevention and early detection of depression and suicide among the youth in Kenya Role: PRINCIPAL INVESTIGATOR: DR. FAITH MUENI MUSYOKA PERIOD: 1 st OCTOBER 2023- 30 th October 2025 AWARDING BODY: Canadian International Development Research Centre (IDRC) and the United Kingdom Foreign Commonwealth Development Office (FCDO), The African Centre for Technology Studies (ACTS)& The National Research Fund (NRF), Kenya AMOUNT: USD 37,069
	Title	Application of Generative Artificial Intelligence to Enhance Practical Teaching of Computer Programming Using a High-Level Language. Role: Co-Principal Investigator PERIOD: 1 st March, 2024- 28 th February 2025 AWARDING BODY: Kenya Education Network (KENET), Kenya AMOUNT: KES. 1,500,000



Research Interests

- Internet of Things
- Data Governance and Ethical AI
- Machine Learning
- Health Informatics
- Software Quality Metrics

References

Upon Request